Task 1 (AWS Cafe Challenge Portfolio)

Challenge 2 - Creating a Dynamic Website for the Café

Academic Year	Module	Assessment Number	Assessment Type
2026	5CS048 - Cloud Systems	Task 1 Challenge 2	Portfolio

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Challenge (Cafe) lab: Creating a Dynamic Website for the Café

A business request for the café: Launching a dynamic website with online ordering capability (challenge 2)

Task 1: Analyzing the existing EC2 instance (Lab IDE)

Show a screenshot of the EC2 Instances console displaying the Lab IDE instance details (including Instance ID, Public IPv4 address, VPC/subnet placement, security group rules, and IAM role if attached).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
Lab IDE	i-043de1d79cdc29930	Running	t3.medium	Initializing	View alarms +	us-east-1a	ec2-3-227-133-109

Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security group name	Key name	Launch time
3.227.133.109	3.227.133.109	–	enabled	Lab IDE SG	–	2026/03/30 07:33 GMT+5:45

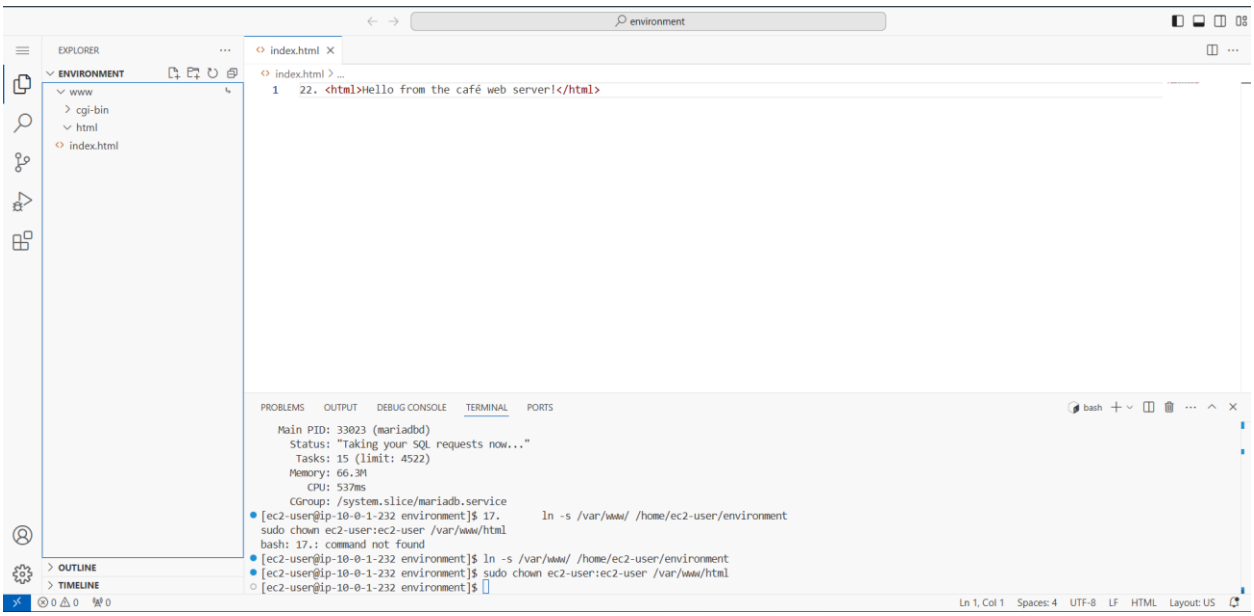
Platfor...	Managed	Operator
Linux/UNIX	false	–

Task 2: Connecting to VS Code IDE and configuring the LAMP stack environment

Show a screenshot of the VS Code IDE open in your browser with:

- **The bash terminal visible**
- **The /var/www/html directory (or symlink) in the file explorer**

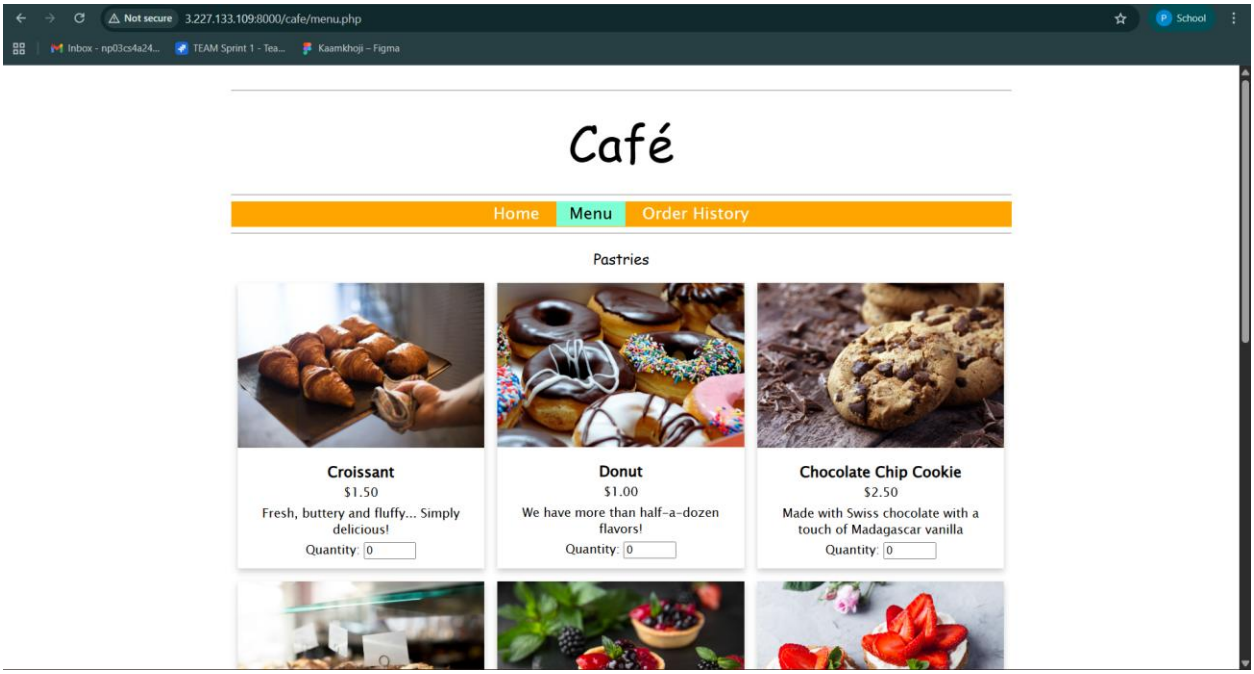
- **The test index.html file created and its content visible ("Hello from the café web server!")**

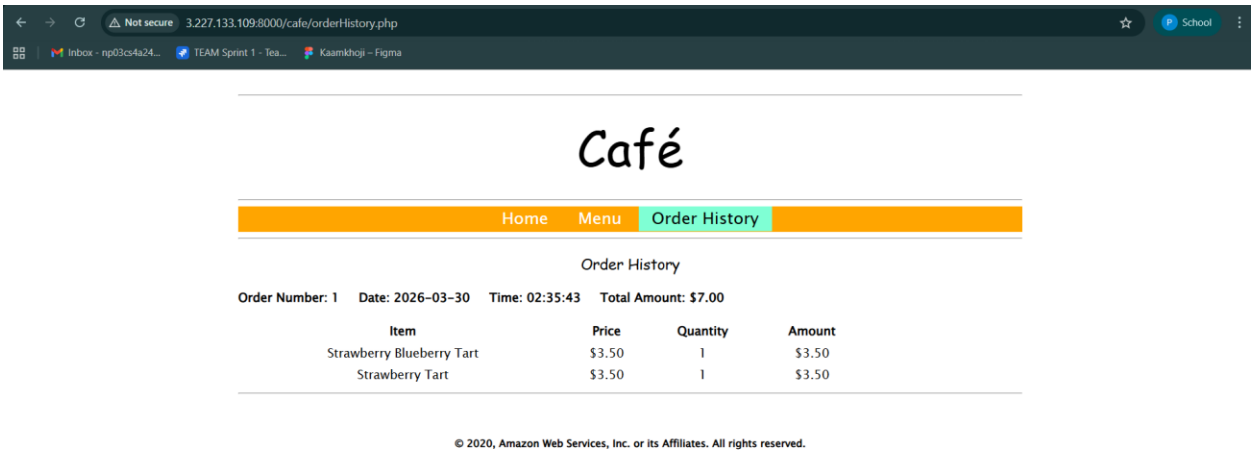


Task 3: Installing and testing the café dynamic web application

Show two screenshots:

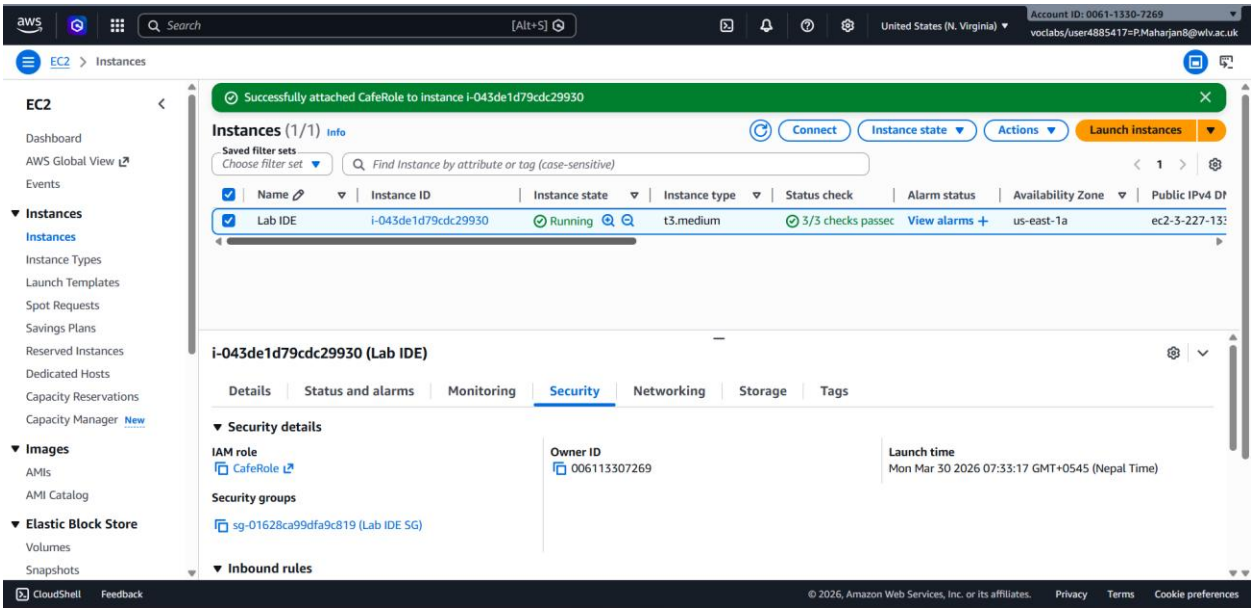
- **One of the browser successfully loading http://<public-ip>:8000/cafe/ displaying the full café menu (products visible).**
- **One of the Order History page after placing at least one test order (showing order details).**





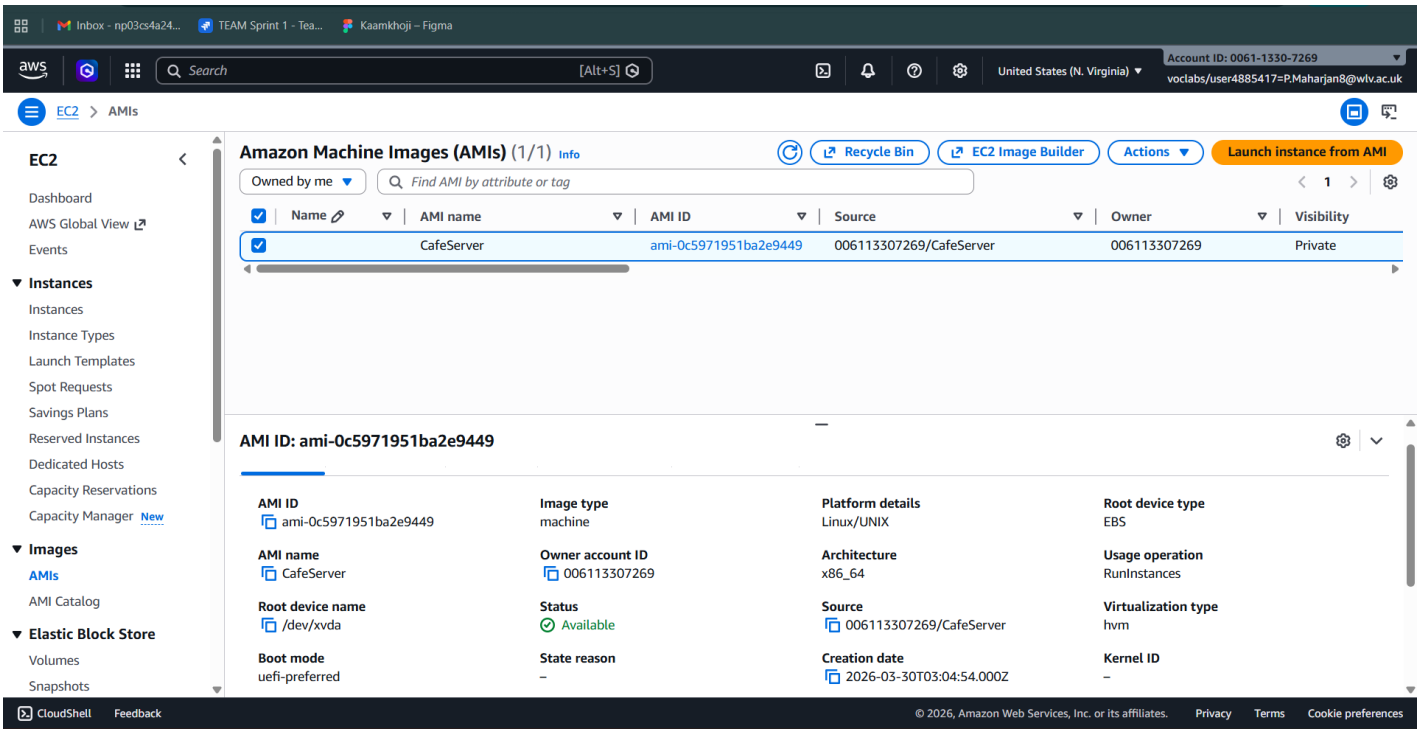
Task 4: Attaching the IAM role and resolving application permission issues

Show a screenshot of the EC2 instance details page with the CafeRole IAM instance profile attached (Security tab or Instance details → IAM role section).



Task 5: Creating an AMI of the development instance

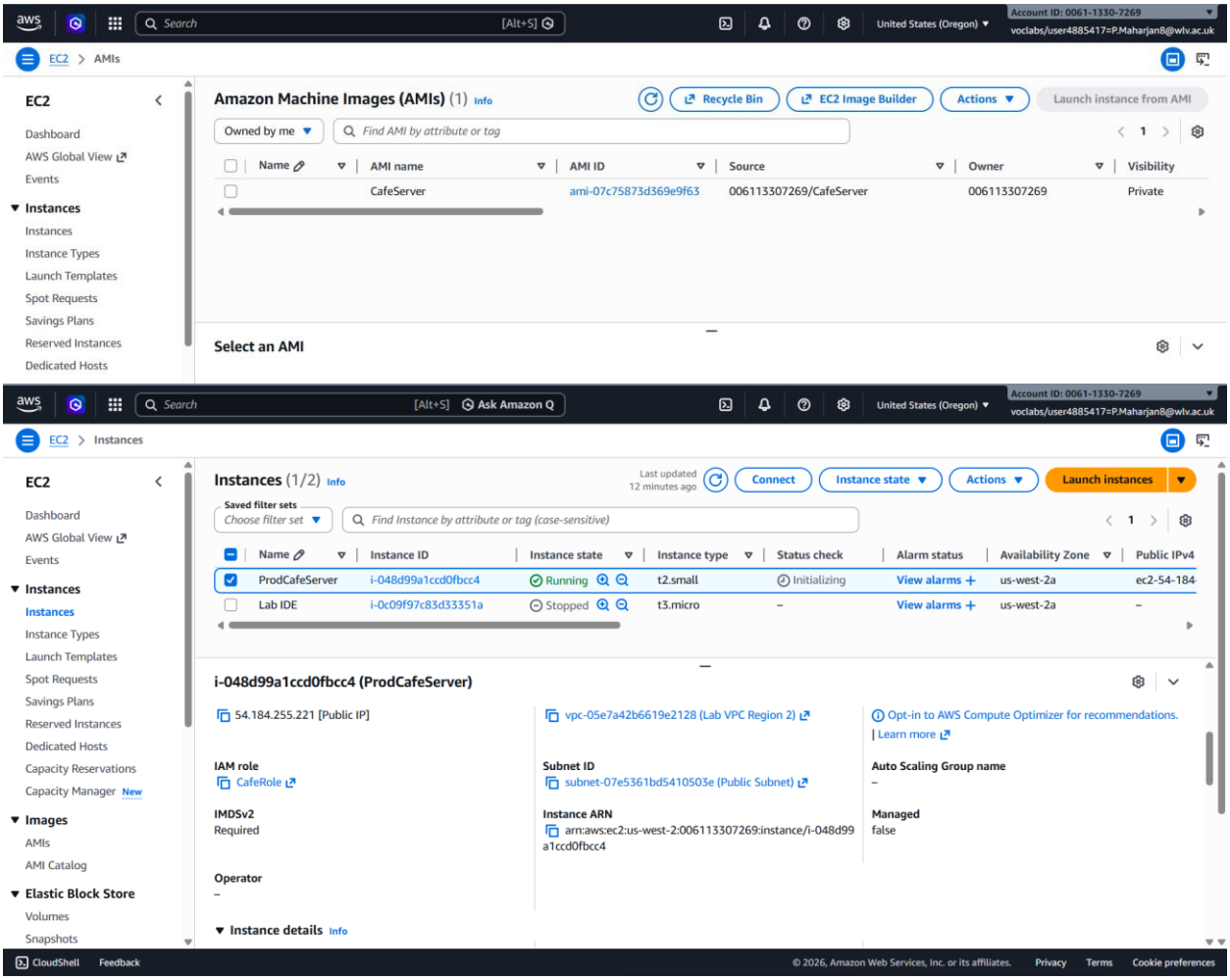
Show a screenshot of the AMIs section in the EC2 console displaying the CafeServer AMI you created (status Available, creation date visible).



Task 6: Copying the AMI to another Region and launching the production instance

Show two screenshots:

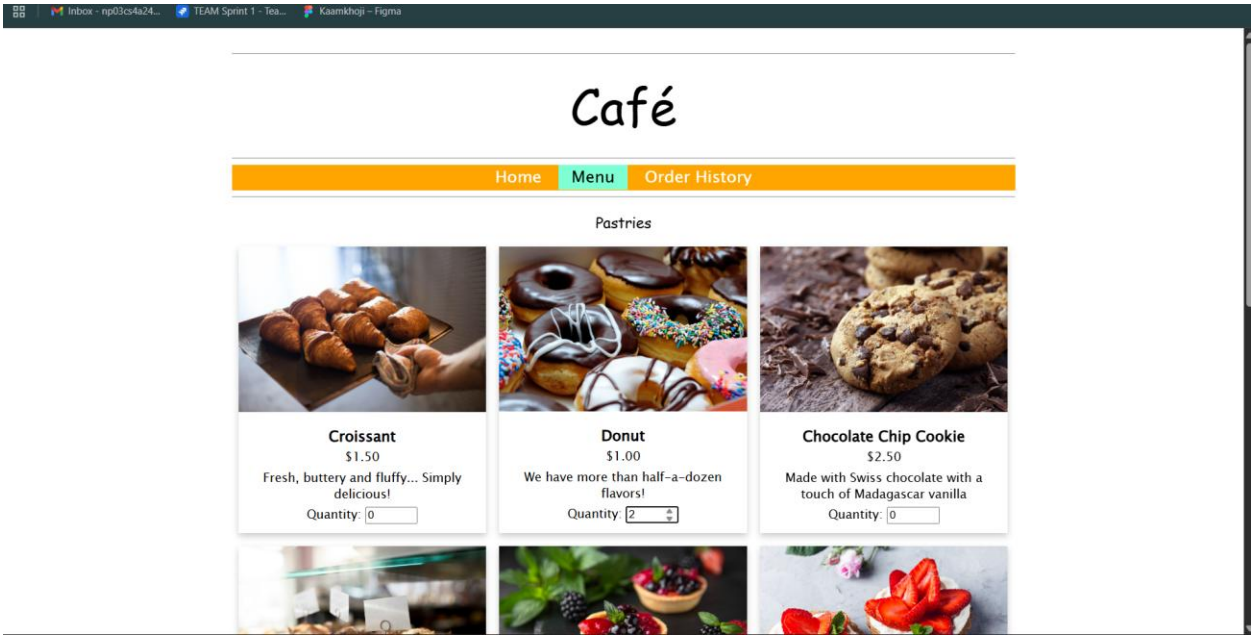
- **The AMI copied / shared / replicated in the Oregon (us-west-2) Region (AMIs list showing CafeServer).**
- **The ProdCafeServer EC2 instance running in us-west-2 (showing Public IPv4 address/DNS, instance state running, security group cafeSG with ports 22 & 8000 open, and CafeRole attached).**



Task 7: Verifying the production website in the second Region

Show a screenshot of the browser loading http://<public-ip-us-west-2>:8000/cafe/ in

the Oregon Region, displaying the full café menu (proof the dynamic application is working).



Description

Write in your own wording and clearly cover all of the following (about 250 to 450 words):

1. what you built and what you did in that lab
2. one security control you used and why it matters
3. one performance or reliability feature that is present and how it helps
4. one issue you had and how you fixed it (or one likely issue and how you would fix it)

Ans :

In this lab, I built and deployed a dynamic café website using an Amazon EC2 instance. Initially, I analyzed the existing EC2 instance and connected to it through the VS Code IDE. I then configured a

LAMP stack environment by setting up the Apache web server, PHP, and a MariaDB database. After confirming that the web server was accessible through the public IP address, I installed the café web application, which allows users to browse the menu and place online orders. The application also integrates with AWS Secrets Manager to securely retrieve database credentials. Once the application was working correctly, I tested it by placing orders and verifying them in the Order History page. Finally, I created an Amazon Machine Image (AMI) from the development instance and launched a production instance in another AWS Region (Oregon), ensuring the application runs consistently across environments.

One important security control used in this lab was the IAM role (CafeRole) attached to the EC2 instance. This role allows the application to securely access sensitive data stored in AWS Secrets Manager without hardcoding credentials in the code. This is important because it reduces the risk of exposing sensitive information such as database passwords and improves overall system security.
A key reliability feature implemented in this setup is the use of multiple AWS Regions for deployment. By creating an AMI and launching a second instance in another region, the system gains redundancy and disaster recovery capability. If one region becomes unavailable, the application can still run from another region, ensuring continuous service availability for users.

One issue I encountered during the lab was that the café website did not load properly even though the web server and database were functioning. This was due to missing permissions for accessing AWS services. I resolved this issue by attaching the required IAM role (CafeRole) to the EC2 instance, which granted the necessary permissions for the application to retrieve secrets and function correctly. Alternatively, such issues can also be troubleshooted by checking security groups, service permissions, and application logs.